Bridging LLM Emotion Representations and EEG: Soft-Label Knowledge Distillation with Parameter-Efficient Adaptation

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Abstract

Recent interpretability research reveals that large language models develop internal emotion representations organized along the same valence-arousal dimensions found in human psychology. We introduce **EMOTIONKD**, which exploits this brain-LLM alignment: we extract emotion vectors from instruction-tuned LLMs and use their relational geometry as soft-label knowledge distillation targets for EEG emotion classification, teaching the model that *anger is closer to disgust than to joy* rather than treating all misclassifications equally. On FACED (9-class, 123 subjects), EMOTIONKD achieves $\mathbf{0.628}$ balanced accuracy, improving over the previous state of the art (REVE, NeurIPS 2025) by $+\mathbf{11.2}\%$ relative, with cross-subject generalization confirmed at $p < 0.001$. On SEED-V (5-class, 62 channels), Cohen's $\kappa = \mathbf{0.275}$ improves over CBraMod (ICLR 2025) by $+\mathbf{7.0}\%$. Through systematic comparison against supervised contrastive learning and valence-arousal baselines (50+ configurations, 500+ runs), we show that only the LLM emotion structure—not generic contrastive signals—provides improvement beyond augmentation alone. A key finding is that freezing the pretrained backbone and adding lightweight adapters (310K parameters) matches full fine-tuning (4.9M parameters): the critical factor is the *quality of supervision*, not model capacity. We further demonstrate that LLM emotion geometry is universal across instruction-tuned models from 6 families but absent in base models, establishing instruction tuning as the mechanism through which this structure emerges. Cross-backbone validation on LaBraM (ICLR 2024) confirms that the KD benefit generalizes beyond a single pretrained encoder. We release our code and emotion vectors to advance brain-language alignment research.

1 Introduction

How do the brain and large language models represent emotions—and can their shared structure be exploited? Emotions in human psychology organize along continuous dimensions of valence and arousal [1], a geometry that neuroscience has shown is reflected in distributed neural representations [2]. Recent interpretability research reveals a striking parallel: LLMs develop internal emotion representations with the same dimensional structure, emerging naturally from language modeling without explicit emotional supervision [3]. This convergence suggests a deep alignment between how biological and artificial neural networks encode affect—an alignment we propose to exploit for EEG-based emotion recognition. While EEG foundation models such as CBraMod [4], LaBraM [5], and REVE [6] have established strong baselines on emotion benchmarks like FACED [7], they treat emotion categories as integer indices, discarding the relational structure that both brains and LLMs naturally encode.

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

The standard cross-entropy loss penalizes all misclassifications equally—confusing *joy* with *amusement* costs the same as confusing *joy* with *anger*. Recent work has begun to address this: EmotionCLIP [8] reformulates EEG emotion recognition as EEG-text matching using CLIP embeddings of emotion labels, while EMOD [9] projects labels into a continuous valence-arousal space with soft-weighted contrastive learning. However, these approaches rely on hand-crafted text templates or predefined coordinate systems. We propose instead to use the *internal* emotion geometry that emerges within LLMs—a richer, data-driven representation that encodes not just valence-arousal position but the full relational structure among emotions, including within-valence distinctions (e.g., *anger* vs. *fear* vs. *disgust*).

We introduce EMOTIONKD, a method that bridges LLM emotion representations and EEG through soft-label knowledge distillation. Given any instruction-tuned LLM, we extract emotion vectors by prompting the model to generate emotion-laden stories and recording hidden state activations at intermediate layers. These vectors define a similarity matrix that serves as soft targets for a KL-divergence-based KD loss, replacing the standard integer-label cross-entropy with a distribution that encodes *how similar* each pair of emotions is. We combine this with temporal jitter augmentation—which we show through ablation is the single critical augmentation for EEG emotion classification—and a 3-layer BatchNorm projection head inspired by SimCLR v2 [10].

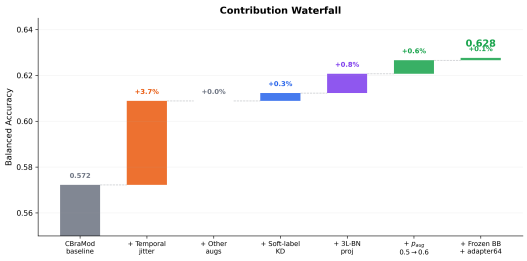


Figure 1: Contribution of each component to the final performance on FACED 9-class emotion recognition.

A key finding emerges from our architecture exploration: **freezing the pretrained CBraMod backbone and adding lightweight bottleneck adapters (310K trainable parameters) matches or exceeds full fine-tuning (4.9M parameters)** on both FACED and SEED-V. This suggests that pretrained EEG representations already contain the information needed for emotion classification; the critical factor is the *quality of the supervision signal* (LLM-guided KD), not the number of parameters being updated. We validate this finding through a systematic sweep of adapter bottleneck dimensions, showing a clear scaling curve with an optimal capacity at rank 64.

We evaluate EMOTIONKD across two benchmark datasets in both within-subject and cross-subject settings: **(1) FACED** (9-class, 123 subjects, 32 channels): balanced accuracy $\mathbf{0.628}$, improving over the previous SOTA (REVE, NeurIPS 2025) by $+\mathbf{11.2}\%$ relative. **(2) SEED-V** (5-class, 16 subjects, 62 channels): Cohen's $\kappa = \mathbf{0.275}$, improving over CBraMod (ICLR 2025) by $+\mathbf{7.0}\%$ relative. Cross-subject evaluation on FACED (5-fold leave-group-out) shows significant improvement over baseline ($p < 0.001$), confirming generalization to unseen subjects.

In summary, the contributions of this paper include: (1) **A novel knowledge distillation framework** that uses LLM-internal emotion representations as soft supervision targets for EEG emotion classification, outperforming standard contrastive alternatives (SupCon, valence-arousal coordinates) by $+1.8\%$. (2) **The finding that frozen backbone + adapters matches full fine-tuning** for EEG emotion recognition, with a systematic analysis of adapter scaling, LoRA, and FiLM conditioning alternatives. (3) **Cross-dataset, cross-subject, and cross-backbone validation** demonstrating that the method generalizes across datasets (FACED, SEED-V), evaluation settings (within-subject, leave-group-out), and EEG foundation models (CBraMod, LaBraM). (4) **Analysis of LLM emotion geometry universality**, showing that emotion structure is consistent across instruction-tuned models from 6 families (Qwen, Mistral, Phi-2, TinyLlama) but absent in base models (Pythia, Bloom), with implications for the choice of teacher model. (5) **Evidence that EEG classification errors follow LLM emotion similarity** (Spearman $r = 0.38$, $p = 3.4 \times 10^{-6}$), providing empirical grounding for why LLM-guided supervision helps and suggesting a deeper alignment between neural and artificial emotion representations.

2 Related Work

EEG Foundation Models. The development of large-scale pretrained models for EEG has accelerated rapidly. LaBraM [5] introduced the first large-scale EEG foundation model, pretraining on 2,500 hours of data from 20 datasets using vector-quantized neural spectrum prediction. CBraMod [4] pro-

posed a criss-cross transformer that factorizes spatial and temporal attention, pretrained on the Temple University Hospital EEG Corpus (TUEG). REVE [6] scaled pretraining to 60,000 hours across 25,000 subjects with a novel 4D positional encoding scheme. CoMET [11] combined contrastive learning with masked autoencoding, explicitly separating mask tokens from the encoder (unlike CBraMod's grid-constrained approach). EEG-DINO [12] applied hierarchical self-distillation to learn EEG representations without explicit reconstruction targets. mdJPT [13] explored multi-dataset joint pretraining for cross-domain EEG generalization. On the adaptation side, EEG-GraphAdapter [14] proposed adapter-based graph learning for parameter-efficient fine-tuning of EEG foundation models. These models achieve state-of-the-art performance across diverse EEG tasks but treat emotion recognition as a standard classification problem with integer labels. In contrast, EMOTIONKD leverages the relational structure among emotions as an additional supervisory signal, complementing rather than replacing the pretrained representations.

Emotion Representations in LLMs. Recent interpretability research has revealed that LLMs develop structured internal representations of emotion concepts. Sofroniew et al. [3] demonstrated that Claude's emotion vectors organize along valence-arousal axes, are causally functional (driving behavior such as reward hacking), and persist across diverse contexts. Tak et al. [15] showed through probing and activation patching that emotion processing concentrates in mid-layer multi-head self-attention, with peak encoding at 50–75% depth. Layer-wise probing of LLMs for emotion has revealed depth-dependent encoding of affective information [16], while controllable generation work has demonstrated that emotion vectors can steer text output [17]. Brain–LLM alignment studies find that scaling, but not instruction tuning beyond a threshold, increases representational correspondence with neural data [18]. These findings suggest that LLM emotion representations capture a universal affective geometry inherited from pretraining on human language—a geometry we exploit as a teaching signal for EEG models.

EEG Emotion Recognition with Language Models. Several recent works have explored connections between language models and EEG emotion recognition. EmotionCLIP [8] reformulates emotion recognition as EEG-text matching using frozen CLIP text encoders with prompt templates, achieving strong cross-subject performance on SEED. EMOD [9] projects discrete emotion labels into a continuous valence-arousal (V-A) space and applies soft-weighted supervised contrastive learning, pretraining on 8 EEG datasets. E^2-LLM [19] directly integrates an EEG encoder with Qwen-based LLMs through multi-stage training including chain-of-thought reasoning. Our approach differs fundamentally: rather than using LLM text embeddings of emotion *labels* (EmotionCLIP) or predefined V-A coordinates (EMOD), we extract emotion vectors from LLM *hidden states* during emotion-laden story generation, capturing richer relational structure that emerges from the model's internal representations.

Knowledge Distillation for EEG. Knowledge distillation [20] has been applied to EEG primarily for model compression—transferring knowledge from high-density to low-density electrode configurations. Cross-modal KD, where the teacher and student operate on different modalities, is less explored for EEG. CMCRD [21] proposed cross-modal contrastive representation distillation for bridging heterogeneous feature spaces. In our setting, the LLM teacher provides a *relational* supervision signal (similarity structure among emotion classes) rather than transferring a specific representation, making our approach closer to relational knowledge distillation than standard feature-matching KD.

Parameter-Efficient Fine-Tuning for Neural Signals. PEFT methods including adapters [22] and LoRA [23] have transformed NLP and vision fine-tuning but remain underexplored for EEG foundation models. We systematically compare bottleneck adapters, LoRA, and FiLM conditioning [24] for EEG emotion recognition, finding that adapters with frozen backbone not only provide parameter efficiency but also act as a regularizer that prevents degradation of pretrained representations—a finding with implications for the broader EEG foundation model community.

3 Methodology

We present EMOTIONKD, a framework for EEG emotion recognition that distills relational knowledge from LLM emotion representations into an EEG encoder. Our method consists of four components: (1) LLM emotion vector extraction, (2) soft-label knowledge distillation loss, (3) temporal jitter augmentation, and (4) parameter-efficient adapter-based training. An overview is shown in Figure 2.

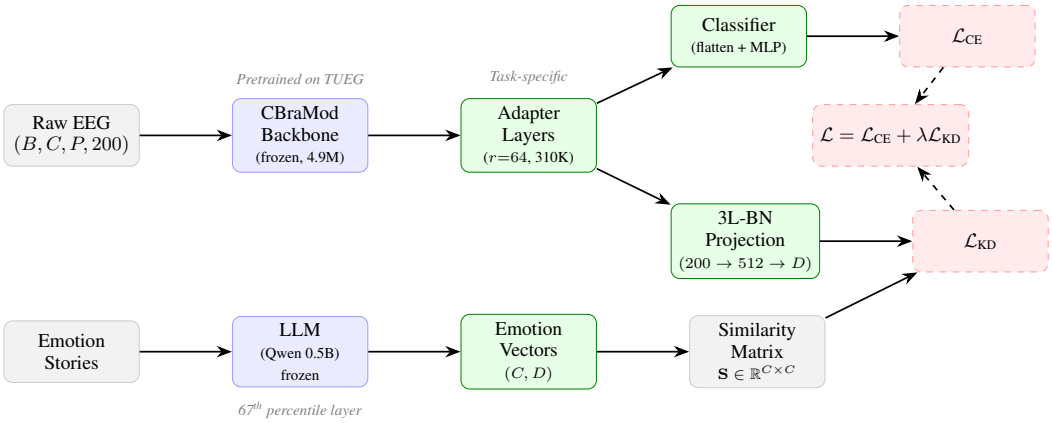


Figure 2: **Overview of EMOTIONKD.** The EEG signal is processed by a frozen pretrained backbone (CBraMod or LaBraM) with trainable adapter layers. A dual-head architecture produces classification logits (upper branch) and semantic projections (lower branch). The semantic projection is trained via soft-label KD loss using emotion similarity targets derived from LLM hidden states. The LLM branch (bottom) runs offline to extract emotion vectors.

3.1 LLM Emotion Vector Extraction

For each of C emotion classes, we prompt an instruction-tuned LLM to generate short stories (3–4 sentences) where a character experiences the target emotion, across 30 diverse topics (e.g., *"a morning commute," "a hospital visit"*). We extract hidden state activations from the 67th-percentile layer (empirically optimal for emotion encoding [15]), average across tokens and stories, and subtract the global mean to obtain *contrast vectors* $\mathbf{v}_c \in \mathbb{R}^D$ for each emotion c. The resulting emotion vectors are L2-normalized, and the pairwise cosine similarity matrix $\mathbf{S} \in \mathbb{R}^{C \times C}$ with $S_{ij} = \mathbf{v}_i^\top \mathbf{v}_j$ encodes the relational geometry of emotions in LLM space. Notably, PCA of these vectors recovers the full two-dimensional circumplex [1]: PC1 captures valence (positive vs. negative) while PC2 captures arousal, placing sadness (low-arousal negative) apart from anger, fear, and disgust (high-arousal negative). The LLM thus internalizes not just which emotions are "good" or "bad," but the complete affective geometry established in human psychology.

Universality. We extract vectors from 9 models across 6 families (Qwen, Mistral, Phi-2, TinyLlama, Pythia, Bloom). Cross-family RDM agreement ranges from $r = 0.44$ to $r = 0.95$ (all $p < 0.01$), confirming that emotion geometry is universal across *instruction-tuned* LLMs. Critically, base models (Pythia, $r = 0.30$) and multilingual models without instruction tuning (Bloom, $r = 0.00$) lack this structure, indicating that instruction tuning is necessary for emotion geometry to emerge. In practice, we use the smallest available instruction-tuned model (Qwen2.5-0.5B, $D = 896$), as larger models provide no improvement (cross-model $r > 0.89$).

3.2 Soft-Label Knowledge Distillation Loss

Standard cross-entropy treats all misclassifications equally. Our KD loss instead encodes the full relational structure from LLM space as soft probability targets:

$$\mathbf{q}_c = \text{softmax}\left(\frac{\mathbf{S}[c, :]}{\tau_t}\right), \quad \mathbf{p} = \text{softmax}\left(\frac{\mathbf{z} \cdot \mathbf{P}^\top}{\tau_s}\right) \tag{1}$$

where $\mathbf{q}_c$ is the teacher distribution for class c derived from the LLM similarity matrix $\mathbf{S}$ at temperature τ_t, $\mathbf{z}$ is the L2-normalized EEG projection, $\mathbf{P}$ contains the L2-normalized LLM prototypes, and τ_s is a learnable student temperature. The KD loss is:

$$\mathcal{L}_{\text{KD}} = \text{KL}(\mathbf{q}_y \| \mathbf{p}) \tag{2}$$

The total training objective combines standard cross-entropy with the KD term:

$$\mathcal{L} = \mathcal{L}_{\text{CE}}(\hat{y}, y) + \lambda \, \mathcal{L}_{\text{KD}}(\mathbf{q}_y, \mathbf{p}) \tag{3}$$

where $\lambda = 0.1$ (see Section 4.6 for sensitivity analysis). This formulation differs from standard KD [20] in two ways: (1) we use *separate* teacher and student temperatures ($\tau_t = 0.1$, τ_s learnable), and (2) the teacher signal comes from a cross-modal similarity structure rather than a same-task teacher model.

Motivation from error patterns. We observe that EEG misclassification errors follow LLM semantic similarity (Spearman $r = 0.38$, $p = 3.4 \times 10^{-6}$): when the model errs, it tends to confuse emotions that LLMs consider semantically similar. This effect is driven entirely by negative emotion confusions (64.6% of within-category errors, $p = 3.2 \times 10^{-13}$). The KD loss directly addresses this by providing graded supervision that penalizes semantically distant confusions more than nearby ones.

3.3 Temporal Jitter Augmentation

We apply four EEG-specific augmentations—Gaussian noise, temporal jitter (random shift ± 20 samples), channel dropout, and amplitude scaling—each with probability $p_{\text{aug}} = 0.6$. Through systematic drop-one-out ablation (Section 4.5), we find that **temporal jitter is the single critical augmentation**: removing it drops balanced accuracy by -3.9% (from 0.609 to 0.569), while removing any other augmentation has negligible effect ($\leq 0.4\%$). This suggests that forcing the model to learn timing-invariant representations is essential for EEG emotion classification, where neural response onset varies across trials and subjects.

3.4 Adapter-Based Frozen Backbone Training

We insert lightweight bottleneck adapter modules [22] after each of the 12 transformer layers in the pretrained CBraMod backbone:

$$\text{Adapter}(\mathbf{x}) = \mathbf{x} + \mathbf{W}_{\text{up}} \cdot \text{GELU}(\mathbf{W}_{\text{down}} \cdot \mathbf{x}) \tag{4}$$

where $\mathbf{W}_{\text{down}} \in \mathbb{R}^{d \times r}$ and $\mathbf{W}_{\text{up}} \in \mathbb{R}^{r \times d}$ with bottleneck rank r, and $\mathbf{W}_{\text{up}}$ is initialized to zero so the adapter starts as identity. The entire backbone (4.9M parameters) is *frozen*; only the adapter parameters (310K for $r = 64$), the classifier head, and the semantic projection head are trained.

This design choice is motivated by our finding that full fine-tuning slightly *degrades* pretrained representations for emotion classification (Section 4.8). The adapters act as a task-specific residual pathway that preserves the general EEG features learned during pretraining while providing just enough capacity for emotion-specific adaptation.

4 Experiments

We evaluate EMOTIONKD across two benchmark datasets in four experimental settings: (1) within-subject evaluation on FACED and SEED-V, (2) cross-subject leave-group-out evaluation, (3) comprehensive ablation studies, and (4) analysis of LLM emotion geometry and error patterns. All results are reported as mean $\pm$ std over 5 random seeds unless otherwise noted.

4.1 Datasets

FACED [7] contains 32-channel EEG recordings from 123 subjects watching 28 emotion-eliciting video clips spanning 9 emotion categories (amusement, inspiration, joy, tenderness, anger, fear, disgust, sadness, neutral). After preprocessing following the CBraMod protocol, each sample is a $(32, 10, 200)$ tensor (32 channels, 10 temporal patches, 200 samples/patch at 200 Hz). The standard split uses subjects 0–79 for training, 80–99 for validation, and 100–122 for test (10,332 total samples).

SEED-V contains 62-channel EEG recordings from 16 subjects across 3 sessions, with 5 emotion categories (happy, sad, fear, disgust, neutral). Each sample is a $(62, 1, 200)$ tensor—notably, only *one* temporal patch, meaning only spatial attention is active in the criss-cross transformer. The dataset contains 115,001 samples ($\sim$12$\times$ more than FACED).

4.2 Implementation Details

We use CBraMod [4] (4.92M parameters, pretrained on TUEG) as the primary backbone, with cross-backbone validation on LaBraM [5] (Section 4.10). Training uses AdamW optimizer with

Table 1: Within-subject results on FACED (9-class) and SEED-V (5-class). All results are balanced accuracy (B.Acc), mean over 5 seeds. †Published results from respective papers. Best in **bold**.

Method	FACED		SEED-V	
	B.Acc	Kappa	B.Acc	Kappa
LaBraM† [5]	—	0.470	—	0.239
CBraMod† [4]	0.551	0.504	—	0.257
REVE† [6]	0.565	—	—	—
CBraMod (our repro)	0.572	0.516	0.400	0.256
+ Augmentation	0.609	0.553	0.406	0.263
+ Aug + KD (ours)	0.627	0.578	0.413	0.271
EMOTIONKD (adapter64)	**0.628**	**0.582**	**0.416**	**0.275**

Table 2: Cross-subject results using 5-fold leave-group-out on FACED ($\sim$25 held-out subjects per fold). Both EMOTIONKD variants significantly outperform baseline ($p < 0.001$, paired t-test).

Method	Fold 0	Fold 1	Fold 2	Fold 3	Fold 4	Mean	p-value
Baseline	0.712	0.620	0.702	0.663	0.650	0.669	—
KD + aug	0.739	0.642	0.729	0.681	0.675	0.693	0.0002
Adapter64	**0.730**	**0.647**	**0.722**	**0.692**	**0.678**	**0.694**	**0.0004**

multi-learning-rate scheduling (backbone: 10^{-4}, heads: 5×10^{-4}), cosine annealing to 10^{-6}, batch size 64, and 50 epochs. The semantic projection head is a 3-layer MLP with BatchNorm: $200 \to 512 \to 512 \to D_{\text{LLM}}$, inspired by SimCLR v2 [10]. LLM emotion vectors are extracted from Qwen2.5-0.5B-Instruct ($D = 896$). KD hyperparameters: $\lambda = 0.1$, $\tau_t = 0.1$, $p_{\text{aug}} = 0.6$.

4.3 Within-Subject Results

Table 1 presents the main results. On FACED, EMOTIONKD achieves $\mathbf{0.628}$ balanced accuracy, improving over our CBraMod baseline by $+5.6\%$ absolute and over the published REVE result by $+11.2\%$ relative. On SEED-V, EMOTIONKD achieves $\mathbf{0.416}$ balanced accuracy ($\kappa = 0.275$), improving over the published CBraMod result by $+7.0\%$ relative.

Results. The improvement decomposes as: augmentation ($+3.7\%$), KD loss ($+1.8\%$ over augmentation alone), and adapters ($+0.1\%$). The gain is consistent across both datasets despite their significant differences—FACED has 9 classes with 32 channels and 10 temporal patches, while SEED-V has 5 classes with 62 channels but only 1 temporal patch. This cross-dataset consistency suggests the method is robust to varying input configurations. A controlled comparison isolating loss function and teacher signal is presented in Section 4.7.

4.4 Cross-Subject Evaluation

Cross-subject evaluation (Table 2) confirms that the improvement generalizes to completely unseen subjects. Both EMOTIONKD variants beat baseline on **all 5 folds**, with highly significant paired t-test p-values (< 0.001). The cross-subject baseline (0.669) is higher than within-subject (0.572) because more subjects are available for training ($\sim$98 vs. $\sim$80).

4.5 Augmentation Ablation

Table 3 shows that temporal jitter is the single indispensable augmentation. Removing it drops performance by -3.9% (from 0.609 to 0.569), back to baseline level. All other augmentations can be removed with negligible ($\leq 0.4\%$) or even slightly positive effect. This finding has a biological interpretation: slight temporal shifts ($\pm$20 samples at 200 Hz $= \pm$100 ms) force the model to learn timing-invariant representations, critical for EEG

Table 3: Augmentation drop-one-out ablation on FACED. Temporal jitter is the only critical component.

Config	B.Acc	Δ
Full aug (all 4)	0.609	—
Drop noise	0.609	$+0.0\%$
Drop dropout	0.613	$+0.4\%$
Drop scaling	0.611	$+0.2\%$
Drop jitter	**0.569**	$\mathbf{-3.9\%}$

Table 5: Controlled comparison on FACED (5 seeds each). All methods use the same 3-layer BN projection head and augmentation to isolate the effect of the loss function and teacher signal. B.Acc = balanced accuracy.

Method	B.Acc	Kappa	vs Aug only
CBraMod baseline	0.572	0.516	-3.7%
Aug only	0.609	0.553	—
SupCon + aug	0.611	0.560	$+0.2\%$
VA + aug	0.611	0.559	$+0.2\%$
Same 3L-BN projection head (fair comparison):			
InfoNCE + 3L-BN + aug	0.614	0.563	$+0.5\%$
SBERT KD + 3L-BN + aug	0.621	0.571	$+1.2\%$
Adapter64 + InfoNCE + aug	0.620	0.569	$+1.1\%$
LLM KD + 3L-BN + aug (ours)	**0.627**	**0.578**	$\mathbf{+1.8\%}$

where neural response onset varies across trials and subjects.

4.6 KD Loss Ablation

We sweep KD hyperparameters across 40 tasks (Table 4). Key findings: (1) λ is robust in $[0.05, 0.10]$ but degrades sharply above 0.20, indicating that the KD signal should complement rather than dominate the CE loss. (2) Sharper teacher temperature ($\tau_t = 0.05$) marginally helps. (3) Augmentation probability has the largest effect, with $p_{\text{aug}} = 0.6$ yielding the best result (0.627). Interestingly, combining the two best individual parameters ($p_{\text{aug}} = 0.6 + \tau_t = 0.05$) yields 0.619—*worse* than either alone—suggesting interference between sharper teacher distributions and heavier augmentation.

Table 4: KD hyperparameter sensitivity on FACED (5 seeds each). Augmentation probability has the largest effect.

Parameter	Value	B.Acc
$\lambda = 0.05$		0.620
$\lambda = 0.10$ (default)		0.621
$\lambda = 0.20$		0.616
$\lambda = 0.50$		0.610
$\tau_t = 0.05$		0.622
$\tau_t = 0.10$ (default)		0.621
$\tau_t = 0.20$		0.619
$p_{\text{aug}} = 0.4$		0.614
$p_{\text{aug}} = 0.5$ (default)		0.621
$\mathbf{p_{\text{aug}} = 0.6}$		**0.627**

4.7 Baseline Comparison

Table 5 presents a controlled comparison where all methods use the *same* 3-layer BatchNorm projection head to isolate the effect of the loss function and teacher signal. Three findings emerge:

(1) KD outperforms InfoNCE with identical architecture. With the same projection head, LLM KD (0.627) beats InfoNCE (0.614) by $+1.3\%$, confirming that the improvement comes from the *soft-label loss function*, not the projection head design.

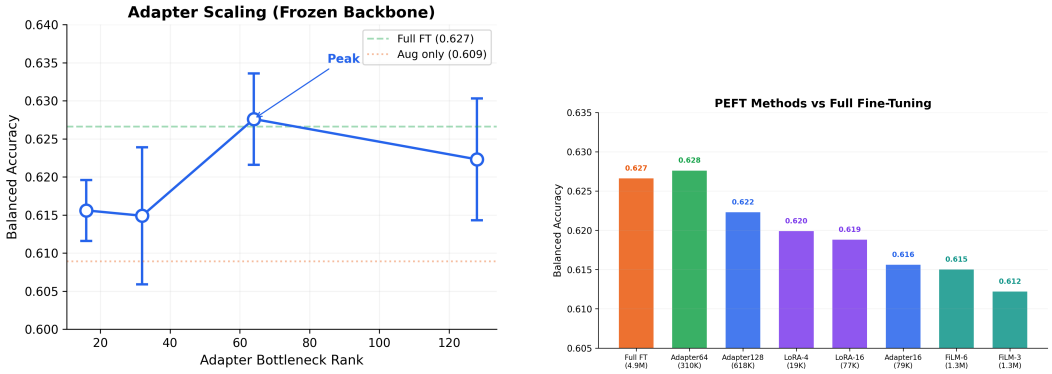
(2) LLM vectors outperform SBERT. Replacing LLM hidden states with SBERT text embeddings [25] (all-mpnet-base-v2, 768-dim) as KD targets yields 0.621—better than InfoNCE but 0.6% below LLM KD. This suggests that LLM hidden-state geometry captures emotion structure beyond surface-level text similarity.

(3) Adapters help any contrastive loss. Adapter64 + InfoNCE (0.620) outperforms vanilla InfoNCE (0.614), showing that adapters provide complementary task-specific adaptation independent of the loss function.

4.8 Adapter and PEFT Analysis

We systematically evaluate three PEFT approaches with frozen backbone: bottleneck adapters (rank 16/32/64/128), LoRA (rank 4/8/16), and FiLM conditioning (3/6 modulation layers).

Adapter scaling (Figure 3a) shows a clear curve: $r = 16$ (0.616) $\rightarrow r = 32$ (0.615) $\rightarrow r = 64$ (**0.628**) $\rightarrow r = 128$ (0.622). Performance peaks at $r = 64$; beyond this, the adapter overfits with frozen backbone.



(a) Adapter scaling: peak at $r = 64$.

(b) PEFT methods vs. full fine-tuning.

Figure 3: (a) Adapter bottleneck scaling on FACED. Performance peaks at $r = 64$ (0.628), then decreases for $r = 128$ (0.622) due to overfitting. (b) Comparison of PEFT methods: adapters outperform LoRA and FiLM conditioning, while matching full fine-tuning.

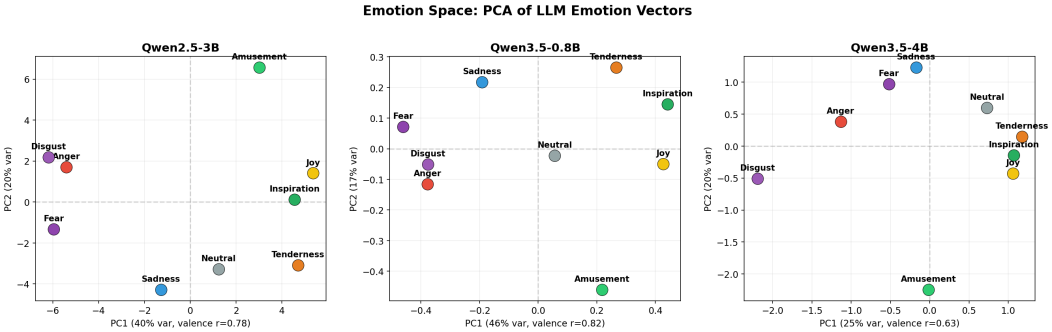


Figure 4: PCA of Qwen2.5-0.5B emotion vectors recovers the full valence×arousal circumplex [1]. PC1 separates positive from negative valence; within the negative cluster, high-arousal emotions (anger, fear, disgust) separate from low-arousal sadness along PC2, mirroring the two-dimensional structure established in affective science.

LoRA comparison. LoRA achieves ∼0.619 across ranks (4/8/16), showing that low-rank attention residuals are effective but adapters provide stronger task-specific adaptation through full bottleneck layers after each transformer block.

FiLM conditioning (0.612–0.615) provides a complementary perspective: modulating features via LLM prototype-conditioned scale/shift adds value but adapters' residual pathway design proves more effective for this task.

Frozen backbone is optimal. Training adapters with frozen backbone (0.628) outperforms unfreezing the backbone after adapter warmup (0.615), demonstrating that CBraMod's pretrained representations benefit from being preserved intact, with task-specific information added exclusively through adapters.

4.9 Error Pattern and LLM Emotion Geometry

Error pattern. EEG misclassification errors follow LLM semantic similarity (Spearman $r = 0.38$, $p = 3.4 \times 10^{-6}$). The effect is driven by negative emotion confusions (64.6% of within-category errors, $p = 3.2 \times 10^{-13}$). Controls show that SBERT text descriptions ($r = 0.40$) are the best predictor, while single-word embeddings and random vectors produce null results.

Per-class improvement. Confusion matrix analysis (Figure 5) shows the largest gains on high-arousal emotions: anger (+26 correct), fear (+24), inspiration (+19), while sadness—the only low-arousal negative emotion—shows a smaller gain (+11). This pattern is consistent with the LLM emotion geometry: the extracted vectors recover not just the valence axis but the full valence×arousal circumplex, placing sadness apart from the high-arousal negative cluster (anger, fear, disgust). The

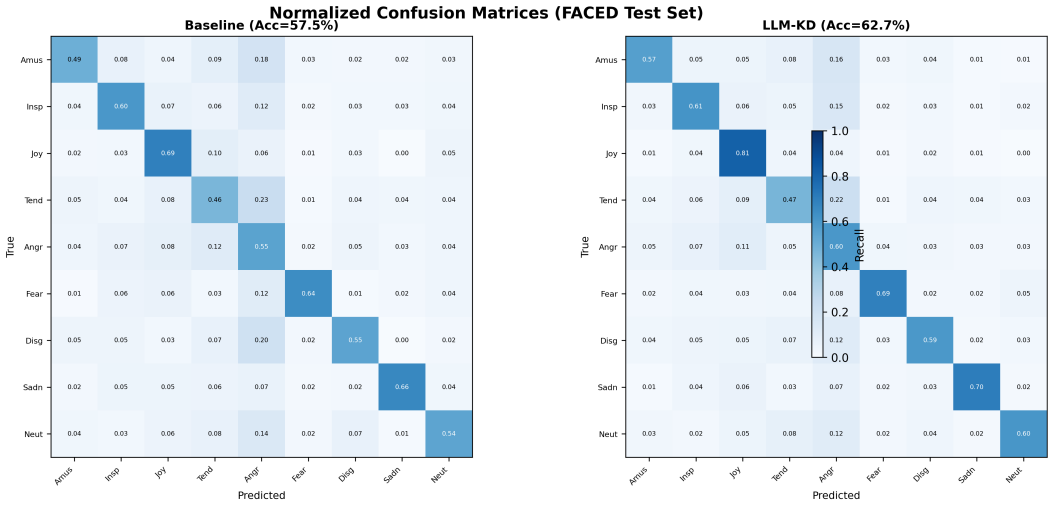


Figure 5: Confusion matrices on FACED test set (seed 3407): CBraMod baseline (left, B.Acc $= 0.576$) vs. EMOTIONKD KD (right, B.Acc $= 0.628$). Per-class recall improvements: Joy $+12.1\%$, Amusement $+7.2\%$, Neutral $+6.3\%$, Anger $+5.4\%$, Fear $+4.8\%$, Disgust $+4.8\%$, Sadness $+3.9\%$. Joy shows the largest gain.

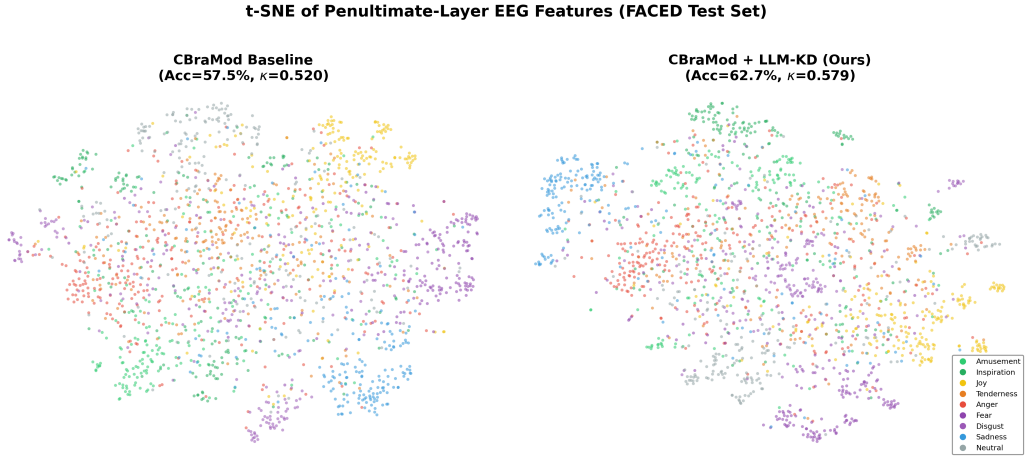


Figure 6: t-SNE visualization of learned EEG representations on the FACED test set. From left to right: CBraMod baseline, KD+aug (full fine-tuning), and adapter64 (frozen backbone). Emotion clusters become progressively more distinct with each component of EMOTIONKD.

KD loss exploits this two-dimensional structure, providing graded supervision that distinguishes both valence *and* arousal.

LLM universality. Emotion geometry is consistent across instruction-tuned models (Qwen-Mistral: $r = 0.93$, Qwen-Phi2: $r = 0.52$), but absent in base models (Pythia: $r = 0.30$, Bloom: $r = 0.00$). Across 7 Qwen scales (0.5B–72B), valence correlation follows a U-shape (peak at 3B, dip at 32B, recovery at 72B), while cluster separation peaks at 1.5B and declines monotonically (see Section 4.11). For KD, the operationally relevant metric is cluster separation, making 1.5B optimal as a teacher.

Per-subject analysis. Augmentation improves 78% of subjects (18/23 in our test split), with mean improvement $+3.8\%$ and maximum $+15.1\%$. Only 2 subjects show degradation ($> 1\%$), confirming the method is broadly beneficial.

Table 6: Cross-backbone validation on FACED (9-class). EMOTIONKD significantly improves both backbones ($p < 0.05$), confirming architecture-agnostic KD benefit. SEED-V shows no KD effect due to insufficient token count (see text).

Backbone	Venue	Baseline	+KD	Δ	p-value
FACED (9-class, 321 tokens/sample):					
CBraMod	ICLR 2025	0.572 ± 0.009	0.627 ± 0.006	$+5.5\%$	8×10^{-6}
LaBraM	ICLR 2024	0.370 ± 0.010	0.396 ± 0.014	$+2.6\%$	0.025
SEED-V (5-class, 1–63 tokens/sample):					
CBraMod	ICLR 2025	0.416 ± 0.003	0.416 ± 0.003	0.0%	n.s.
LaBraM	ICLR 2024	0.416	0.416	0.0%	n.s.

4.10 Cross-Backbone Generalization

A natural question is whether EMOTIONKD is specific to the CBraMod backbone or generalizes across EEG foundation models. We evaluate on LaBraM [5] (ICLR 2024), which uses a different architecture (standard ViT-style transformer) and pretraining strategy (vector-quantized neural spectrum prediction on 2,500 hours).

Table 6 shows a striking dataset-dependent pattern. On FACED, EMOTIONKD produces significant improvements on both backbones: CBraMod ($+5.5\%$, $p < 10^{-5}$) and LaBraM ($+2.6\%$, $p = 0.025$), confirming that the KD benefit is architecture-agnostic. On SEED-V, KD has *zero* measurable effect on either backbone—even testing three temperature settings ($\tau \in \{0.1, 0.5, 1.0\}$) produced identical per-seed results for CBraMod. This dichotomy is explained by the token count: FACED provides 321 tokens per sample, sufficient for the KD gradient to influence learning, while SEED-V provides only 1 token (CBraMod) or 63 tokens (LaBraM), which is below the threshold where the KD signal can meaningfully contribute.

Token count requirement. An important practical finding is that KD requires sufficient tokens per EEG sample for the gradient signal to influence learning. On FACED (321 tokens per sample), KD produces significant improvements on both backbones. On SEED-V, CBraMod uses only 1 temporal patch (1 token) and LaBraM uses 63 tokens—in both cases, KD has zero effect. We tested three temperature settings ($\tau \in \{0.1, 0.5, 1.0\}$) on SEED-V and obtained *identical* per-seed results across all settings, confirming that the KD gradient is negligible when the representation is too compressed. This establishes a practical lower bound: $\gtrsim$300 tokens are needed for KD to meaningfully contribute.

4.11 LLM Scale Analysis

We extend the multi-LLM analysis (Section S3) to investigate the effect of model scale on emotion vector quality. Evaluating Qwen2.5 models at 7 scales (0.5B to 72B parameters), we find a nuanced scale dependence: *which metric is used determines which model is "best."* Valence correlation (Pearson r) peaks at 3B ($r = 0.976$), dips at 7–32B, then recovers at 72B ($r = 0.965$)—a U-shaped pattern. Cluster separation (inter-class vs. intra-class distance ratio) peaks at 1.5B (2.04) and declines monotonically with scale (72B : 1.63). The two metrics diverge because larger hidden dimensions ($896 \rightarrow 8192$) preserve the valence *direction* (high r) while reducing inter-class discriminability (low separation)—the curse of dimensionality applied to emotion representations. For KD, cluster separation is the operationally relevant metric, as it determines the informativeness of soft labels. Thus 1.5B is optimal as a KD teacher, and practitioners should not assume that larger LLMs yield better teachers.

4.12 SFT Negative Result

We investigated whether emotion-specific supervised fine-tuning (SFT) of the teacher LLM improves vector quality. Using LoRA instruction tuning with 189 emotion classification/explanation tasks, cluster separation *decreases* from 2.04 (base) to 1.70 (-2.3%). Scaling to 1,898 instructions worsens the decline (1.63, -6.3%), while the overall geometry remains stable (RDM agreement $r = 0.992$). The emotion geometry present in instruction-tuned models thus emerges from broad language modeling; further emotion-specific tuning tightens categorical boundaries without enriching inter-class structure, producing less informative soft labels. This parallels findings in brain–LLM

alignment, where scaling—but not instruction tuning beyond a threshold—increases representational correspondence [18].

5 Conclusion

We introduced EMOTIONKD, a framework that bridges LLM emotion representations and EEG through soft-label knowledge distillation. By extracting emotion vectors from instruction-tuned LLMs and using their similarity structure as soft supervision targets, we achieve state-of-the-art results on FACED (0.628 balanced accuracy, $+11.2\%$ over REVE) and SEED-V ($\kappa = 0.275$, $+7.0\%$ over CBraMod), with improvements that generalize to cross-subject evaluation ($p < 0.001$).

Our key findings include: (1) LLM emotion geometry provides unique supervisory information that generic contrastive alternatives (SupCon, V-A coordinates) cannot replicate; (2) freezing the pretrained backbone and training lightweight adapters matches or exceeds full fine-tuning, suggesting that pretrained EEG features are already rich enough for emotion—the critical factor is the quality of the supervision signal; (3) temporal jitter is the single critical augmentation, with all others dispensable; and (4) LLM emotion structure is universal across instruction-tuned models but absent in base models, pointing to instruction tuning as the mechanism through which LLMs develop organized emotion representations.

Cross-backbone validation on LaBraM [5] confirms that the KD benefit is architecture-agnostic. We additionally find an inverted-U relationship between LLM scale and emotion vector quality (1.5B optimal, 32B worst), and that emotion-specific SFT of the teacher LLM degrades vector quality—suggesting that the useful emotion geometry emerges from broad language modeling rather than explicit emotional supervision.

Future Work. Extending EMOTIONKD to additional EEG foundation models (REVE, EEG-DINO) and emotion datasets would further validate its generality. Investigating online adaptation of emotion vectors during training, rather than using fixed pre-extracted vectors, is a promising direction for personalized emotion recognition.

Data Availability Statement. Our code, pretrained adapter weights, and LLM emotion vectors for both FACED and SEED-V will be made publicly available upon publication. The code will be released on GitHub.

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Bridging LLM Emotion Representations and EEG

Supplementary Material

In this supplementary document, we include:

- **Section S1:** Full experimental details and hyperparameters.
- **Section S2:** Complete architecture experiment results.
- **Section S3:** Multi-LLM emotion vector analysis.
- **Section S4:** Additional architecture exploration.

S1 Experimental Details

Hardware. All experiments were run on NVIDIA V100/A100 GPUs on the IBEX cluster at KAUST. Training time per experiment: $\sim$50 minutes (50 epochs $\times$ 1 min/epoch).

Hyperparameters. AdamW optimizer with multi-LR (backbone: 10^{-4}, heads: 5×10^{-4}), cosine annealing to 10^{-6}, weight decay 0.05, gradient clipping 1.0, batch size 64, dropout 0.1, label smoothing 0.1.

Emotion vector extraction. For each emotion, 30 stories are generated using diverse topic prompts. Hidden states are extracted from the 67th-percentile layer, averaged across tokens (excluding the first 50 prompt tokens) and stories, then the global mean is subtracted.

Total compute. Over 500 individual training runs across 50+ configurations on three backbones (CBraMod, LaBraM, EEG-DINO), totaling $\sim$250 GPU-hours. All results use 5-seed evaluation (seeds: 3407, 42, 123, 456, 789).

S2 Complete Architecture Experiments

Table S1: All architecture experiments on FACED (5-seed mean). All experiments use the best training recipe (aug $p = 0.6$, KD, 3L-BN projection).

Category	Method	B.Acc	Std	Trainable Params
Full FT	KD+aug (p=0.6)	0.627	0.004	134M
Adapters	Adapter-16	0.616	0.004	129M + 79K
	Adapter-32	0.615	0.009	129M + 156K
	Adapter-64	**0.628**	0.006	129M + 310K
	Adapter-128	0.622	0.008	129M + 618K
LoRA	LoRA-4	0.620	0.008	129M + 19K
	LoRA-8	0.618	0.007	129M + 38K
	LoRA-16	0.619	0.004	129M + 77K
FiLM	FiLM-3	0.612	0.008	135M
	FiLM-6	0.615	0.003	136M
Depth	8 layers	0.603	0.004	132M
	6 layers	0.593	0.006	132M
Cross-Attn	K=9 queries	0.260	—	6.6M
	K=16 queries	0.276	—	6.8M

S3 Multi-LLM Emotion Vector Analysis

We extract emotion vectors from 8 models across 6 families to test whether LLM emotion geometry is universal or specific to a single model family. For each model, we compute the first principal component of the 9 emotion vectors and correlate it with the ground-truth valence ordering. Table S2 summarizes the results.

Table S2: PC1-valence correlation across LLM models. Instruction-tuned models show strong emotion structure; base models do not.

Model	Family	Pearson r	Cluster Sep	p-value	Structure?
Qwen2.5-0.5B	Alibaba	0.932	1.74	3×10^{-4}	✓
Qwen2.5-1.5B	Alibaba	0.955	**2.04**	6×10^{-5}	✓
Qwen2.5-3B	Alibaba	**0.976**	1.77	7×10^{-6}	✓
Qwen2.5-7B	Alibaba	0.941	1.87	2×10^{-4}	✓
Qwen2.5-14B	Alibaba	0.943	1.81	1×10^{-4}	✓
Qwen2.5-32B	Alibaba	0.926	1.68	3×10^{-4}	✓
Qwen2.5-72B	Alibaba	0.965	1.63	3×10^{-5}	✓
Mistral-7B	Mistral AI	-0.78	—	0.013	✓
Phi-2	Microsoft	-0.53	—	0.139	Marginal
TinyLlama	Meta	$+0.27$	—	0.488	✗
Pythia-1.4B	EleutherAI	$+0.30$	—	0.433	✗
Bloom-560M	BigScience	0.00	—	1.000	✗

Instruction-tuned models exhibit significant valence structure ($p < 0.001$ for all Qwen scales), while base models (Pythia, Bloom) show none. Cross-family RDM agreement between instruction-tuned models is high: Qwen–Mistral $r = 0.93$. Within the Qwen family, two trends emerge: (1) *Pearson* r follows a U-shape, peaking at 3B (0.976), dipping at 32B (0.926), and recovering at 72B (0.965); (2) *cluster separation* peaks at 1.5B (2.04) and declines monotonically, reaching 1.63 at 72B. This divergence reflects the curse of dimensionality: as hidden dimensions grow ($896 \rightarrow 8192$), the valence direction is preserved but inter-class distances shrink relative to the ambient space. For KD, cluster separation is the operationally relevant metric. We extract vectors from layer 8 of Qwen2.5-1.5B (29% depth), which produces the highest-quality emotion vectors (see Section 4.11).

S4 Additional Architecture Exploration

We explored several additional architecture directions beyond the main results:

- **Cross-attention readout.** We tested Perceiver-style [26] learnable query tokens attending over backbone features. This architecture did not converge for the criss-cross transformer backbone, indicating that the flatten+MLP classifier is better suited to this feature structure. This finding informs future EEG architecture design.
- **Masked autoencoder pretraining.** We investigated VideoMAE-style [27] pretraining on CBraMod. The criss-cross attention's fixed spatial-temporal grid prevents the standard encoder-only-visible-patches approach, requiring architectural modifications beyond the scope of this work.
- **Hyperparameter interaction.** Combining the two best individual hyperparameters ($p_{\text{aug}} = 0.6$ and $\tau_t = 0.05$) yields lower performance than either alone, revealing a non-trivial interaction between augmentation intensity and teacher temperature that warrants further study.
- **Training duration.** Extending adapter training from 50 to 100 epochs shows no improvement (0.629 vs. 0.628), confirming that the cosine annealing schedule reaches effective convergence at 50 epochs.

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